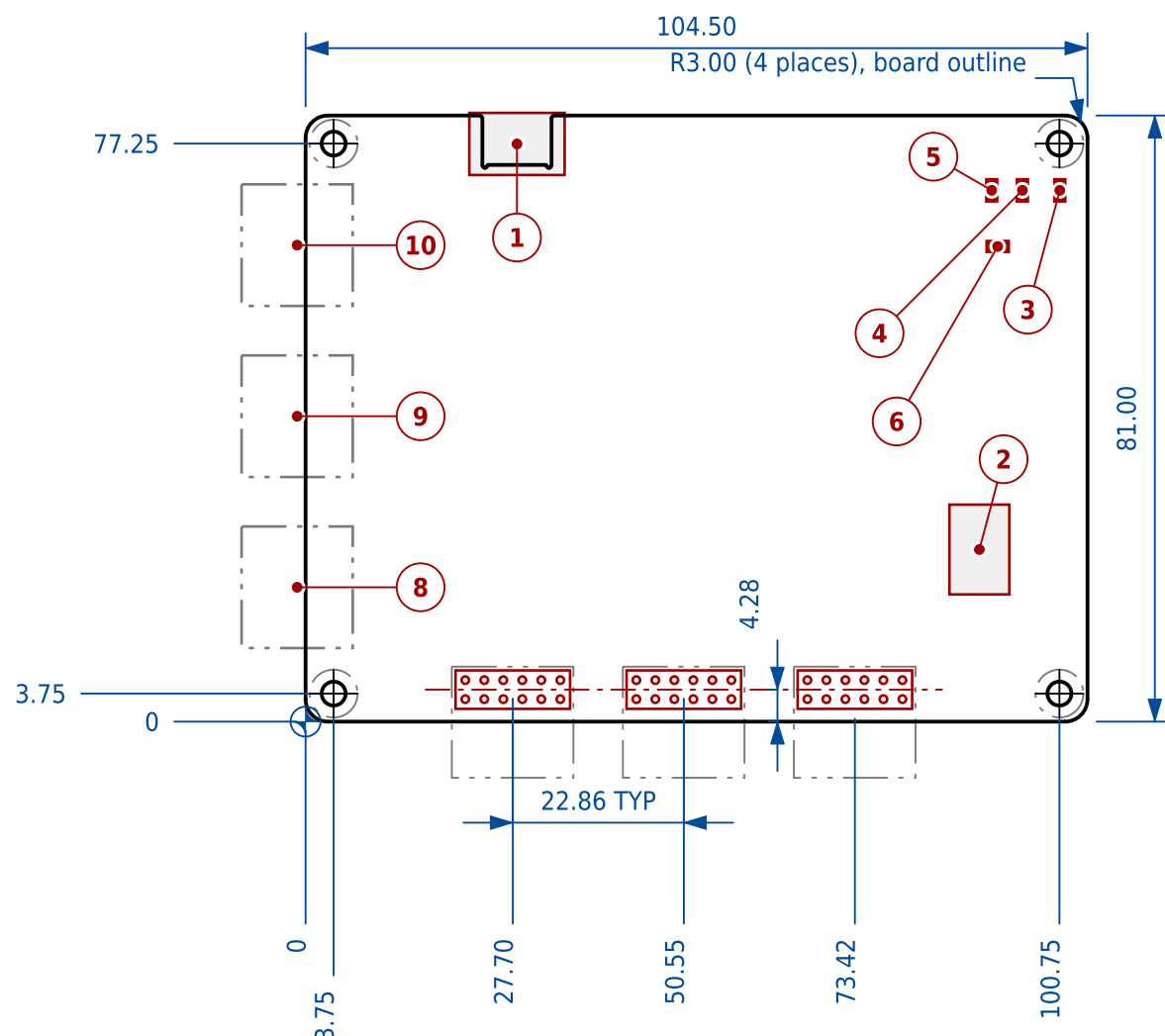




## NOTES

1. All dimensions in millimetres. The datum symbol marks the origin: the board's lower-left corner, X right, Y up, seen from the component side.
2. Geometry is design nominal, read from the KiCad board file.
3. Hole IDs are the board's own reference designators from the KiCad file, not assigned by this drawing, and are in no positional order.
4. Pmod host headers are on a 22.86 mm (0.9 in) pitch, per the Digilent Pmod Interface Specification 1.2.0.
5. This sheet covers revisions 1.2.2 and 1.2.3. They are the same board mechanically: the generator compares outline, holes, Pmod hosts and every feature before merging them. Only the electrical design and the shuttle differ.
6. The Pmod host headers along the lower edge are what a mounting plate registers against. Their 22.86 mm pitch is the same on every revision; their distance from the lower edge and their position along it are not.
7. The chain-double-dot rectangle at each Pmod host is the connector body where it overhangs the edge: what a peripheral and a bracket must clear.
8. Pmod host coordinates are given to three decimals, unlike the rest of this sheet. Rounded to two, adjacent hosts print 22.85 apart and contradict the 22.86 pitch dimensioned here.

## NOTES (continued)

9. Port names differ by family: JA, JB, JC on the adapter and Raspberry Pi sheets, by signal direction on the demo boards, PMOD 1 to 3 on the plate sheets. TT-MP-02 maps them.
10. Board thickness is 1.6 mm nominal. The KiCad file's 1.56252 mm is its stackup sum, not a specified finished thickness.
11. Assembled envelope, connector overhang included, is 104.50 x 81.35 mm against a 104.50 x 81.00 mm board outline. The FEATURE SCHEDULE gives the extents.
12. Upper edge carries a shallow recess for the USB-C shell, contributed by the connector footprint's own edge cuts (fillet radii 0.136 mm, 0.303 mm, 0.364 mm, 0.400 mm).
13. Feature numbers are fixed across this family, so a number means the same part on every sheet. A part this board does not carry still has a row, marked as such.
14. GENERAL TOLERANCE in the title block is a normal board-house figure, not one any source here states.
15. Generated from the listed sources and not checked by a second party: there is no CHECKED field because nobody has signed it.

## SOURCES

1. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ cfdd80d7b - title block: TinyTapeout 4+ Demoboard rev 1.2.2, dated 2024-04-26

## HOLE SCHEDULE

| ID  | X mm   | Y mm  | DIA mm | KEEPOUT mm |
|-----|--------|-------|--------|------------|
| MT1 | 3.75   | 3.75  | 3.20   | 6.40       |
| MT2 | 100.75 | 3.75  | 3.20   | 6.40       |
| MT3 | 3.75   | 77.25 | 3.20   | 6.40       |
| MT4 | 100.75 | 77.25 | 3.20   | 6.40       |

## PMOD HOST HEADERS

| PORT   | REF | EDGE   | CX mm  | CY mm | PIN1 X mm | PIN1 Y mm |
|--------|-----|--------|--------|-------|-----------|-----------|
| INPUT  | J3  | bottom | 27.695 | 4.275 | 34.045    | 5.545     |
| BIDIR  | J5  | bottom | 50.555 | 4.275 | 56.905    | 5.545     |
| OUTPUT | J6  | bottom | 73.415 | 4.275 | 79.765    | 5.545     |

## FEATURE SCHEDULE

| #  | FEATURE                                    | X EXTENT mm      | Y EXTENT mm    |
|----|--------------------------------------------|------------------|----------------|
| 1  | USB-C power / control                      | 21.88 to 34.62   | 73.05 to 81.35 |
| 2  | 7-segment display                          | 86.05 to 94.05   | 17.00 to 29.00 |
| 3  | LED D1                                     | 100.07 to 101.53 | 69.52 to 72.48 |
| 4  | LED D2                                     | 95.07 to 96.53   | 69.52 to 72.48 |
| 5  | LED D3                                     | 90.97 to 92.43   | 69.52 to 72.48 |
| 6  | LED D4                                     | 91.02 to 93.98   | 62.77 to 64.23 |
| 7  | 8-way input DIP switch - not on this board | -                | -              |
| 8  | Side Pmod J12 (not fitted)                 | -8.55 to 6.30    | 9.83 to 26.08  |
| 9  | Side Pmod J13 (not fitted)                 | -8.55 to 6.30    | 32.69 to 48.94 |
| 10 | Side Pmod J14 (not fitted)                 | -8.55 to 6.30    | 55.55 to 71.80 |

## LEGEND

|       |                                      |
|-------|--------------------------------------|
| —     | Board outline                        |
| —     | Component body, scheduled            |
| - - - | Position not fitted on this revision |
| - - - | Pmod connector body, hole keep-out   |
| —>    | Dimension, extension and leader      |

## SOURCES (continued)

2. KiCad board file: <https://github.com/TinyTapeout/tt-demo-pcb> tinytapeout-demo.kicad\_pcb @ a88cbc08b - title block: TinyTapeout 4+ Demoboard rev 1.2.3, dated 2024-06-11
3. Shuttle mapping: [https://docs.google.com/spreadsheets/d/1xD\\_uemuQcpUnXRY4twOrFzF0urJLVWMdqAnrOJLetM/edit?gid=1250963370](https://docs.google.com/spreadsheets/d/1xD_uemuQcpUnXRY4twOrFzF0urJLVWMdqAnrOJLetM/edit?gid=1250963370) - used by: TT04, TT05

## DB 4+ V1.2.2

tinytapeout-demo rev 1.2.2 and 1.2.3

|                  |                  |                                                |       |
|------------------|------------------|------------------------------------------------|-------|
| DRAWN            | VERSION          | UNITS                                          | SCALE |
| generated        | v0.0-83-g296ee10 | mm                                             | 1:1   |
| SIZE             | SHEET            | DRAWING NO                                     | REV   |
| A3               | 1 OF 1           | TT-DB-02                                       | A     |
| MATERIAL         |                  | GENERAL TOLERANCE                              |       |
| PCB, 1.6 nominal |                  | edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08 |       |